

Name: _____

Date: _____

PYTHAGORAS, (APPLYING PYTHAGORAS IN 3D PROBLEMS)

LO: I can apply Pythagoras' theorem to find lengths in three-dimensional shapes.

Pythagoras in 3D

In **3D problems**, we often need to apply Pythagoras' theorem **twice**: once on the base to find a diagonal, then again using that diagonal and the height.

Strategy: Identify the right-angled triangle in the 3D shape. Often this involves finding a base diagonal first, then using it with the vertical height.

Quick examples

●	Cuboid diagonal: $d = (l^2 + w^2 + h^2)$	3D diagonal formula
●	Space diagonal of 3, 4, 12 cuboid $= (9+16+144) = 13$	apply twice or use formula
●	Pyramid: slant height uses base half-diagonal and vertical height	two-step approach
●	3D diagonal = $l + w + h$	cannot add lengths directly

Key idea

In 3D, apply Pythagoras twice or use the 3D formula:
diagonal = $(l^2 + w^2 + h^2)$.

$$d = (\boxed{l^2 + w^2} + \boxed{h^2})$$

First find base diagonal, then combine with height

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - cuboid space diagonal

Find the space diagonal of a cuboid 3 cm by 4 cm by 12 cm.

$$\text{Base diagonal: } (3^2 + 4^2) = (9 + 16) = 25 = 5$$

Now use base diagonal (5) and height (12)

$$\text{Space diagonal} = (5^2 + 12^2) = (25 + 144) = 169 = 13 \text{ cm}$$

13 cm

Apply Pythagoras twice: once for the base, once with the height.

Example 2 - using the 3D formula

Find the space diagonal of a cuboid 2 cm by 6 cm by 9 cm.

$$\begin{aligned} d &= (2^2 + 6^2 + 9^2) \\ d &= (4 + 36 + 81) = 121 \\ d &= 11 \text{ cm} \end{aligned}$$

11 cm

The 3D formula combines both Pythagoras steps into one.

Example 3 - pyramid slant height

A square-based pyramid has base side 6 cm and vertical height 4 cm. Find the slant height.

Half the base = 3 cm (distance from centre to midpoint of base edge)

Right triangle: base 3, height 4

$$\text{Slant} = (3^2 + 4^2) = (9 + 16) = 25 = 5 \text{ cm}$$

5 cm

The slant height runs from apex to the midpoint of a base edge.

Example 4 - cone

A cone has radius 5 cm and slant height 13 cm. Find the vertical height.

Right triangle: radius 5, slant 13 (hypotenuse)

$$\begin{aligned} h^2 &= 13^2 - 5^2 = 169 - 25 = 144 \\ h &= \sqrt{144} = 12 \text{ cm} \end{aligned}$$

12 cm

The slant height is the hypotenuse; subtract to find the vertical height.

YOUR TURN – PRACTICE (deliberate practice)

1. Cuboid space diagonals

Find the space diagonal of each cuboid.

- a) $1\text{ cm} \times 2\text{ cm} \times 2\text{ cm}$
- b) $2\text{ cm} \times 3\text{ cm} \times 6\text{ cm}$
- c) $4\text{ cm} \times 4\text{ cm} \times 7\text{ cm}$
- d) $6\text{ cm} \times 6\text{ cm} \times 7\text{ cm}$

2. Cube diagonals

Find the space diagonal of each cube (side length given).

- a) Side = 1 cm
- b) Side = 4 cm
- c) Side = 5 cm
- d) Side = 10 cm

3. Pyramids

Find the slant height of each square-based pyramid.

- a) Base 8 cm , height 3 cm
- b) Base 10 cm , height 12 cm
- c) Base 6 cm , height 8 cm
- d) Base 12 cm , height 5 cm

4. Cones and cylinders

Find the missing measurement.

- a) Cone: radius 3 , slant height 5 . Vertical height?
- b) Cone: radius 8 , vertical height 15 . Slant height?
- c) Cone: slant 10 , height 6 . Radius?
- d) Cylinder: radius 3 , height 4 . Diagonal across?

5. Mixed 3D problems

Solve each problem. Give answers to 1 d.p. where necessary.

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) A cuboid space diagonal requires Pythagoras applied twice ☐

Correct answer: _____

Reason: _____

b) The space diagonal of a cuboid equals $l + w + h$ ☐

Correct answer: _____

Reason: _____

c) $d = (l^2 + w^2 + h^2)$ gives the space diagonal ☐

Correct answer: _____

Reason: _____

d) A pyramid slant height equals the vertical height ☐

Correct answer: _____

Reason: _____

e) In a cone, the slant height is the hypotenuse ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A room is **6 m long, 4 m wide, and 3 m high**. A spider at one top corner wants to reach the opposite bottom corner. What is the shortest distance it can travel along the surfaces? (Hint: unfold the room.)

Expression: _____

Simplified: _____

b) A square-based pyramid has base side **8 cm** and all slant edges are **9 cm**. Find the vertical height of the pyramid. Then find the distance from the apex to the midpoint of a base edge.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

